$$
\begin{array}{r|l}
x-3 & x^3-7x^2+7x+15 \\
\hline
 & x^2-4x-5 \\
 & x^3-3x^2 \\
\hline
 & -4x^2+7x \\
 & -4x^2+12x \\
\hline
 & -5x+15 \\
 & -5x+15
\end{array}
$$

$\Rightarrow$

$$x^3-7x^2+7x+15=0$$

$$(x-3)(x^2-4x-5)=0$$

$$(x-3)(x-5)(x+1)=0$$

$$x=\boxed{3,\ 5,\ -1}$$